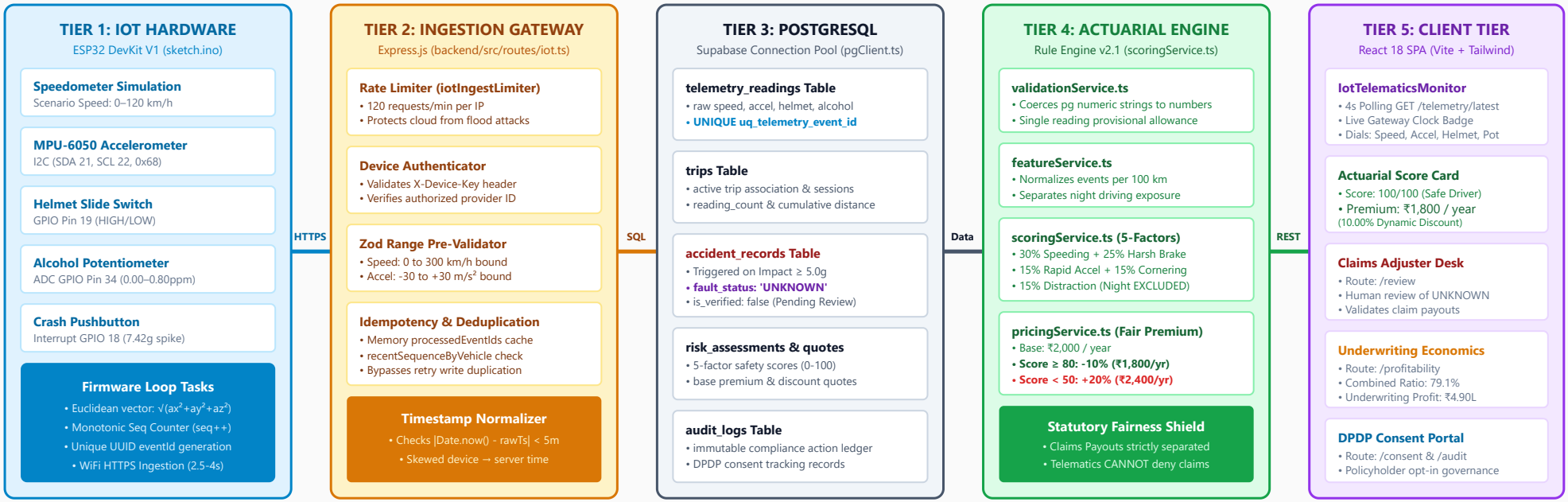
INSURE — Detailed Workflow Architecture Diagram

COMPLETE 5-TIER DATA PIPELINE: EMBEDDED IOT • EDGE GATEWAY • POSTGRESQL • ACTUARIAL ENGINE • REACT CLIENT

Track: **InsurTech & IoT Telematics** • Team: **Techsquad** • SYNAPSE '26
Live Architecture: **insurevsb.vercel.app**



WORKFLOW SUMMARY: HOW ONE SECOND OF DRIVING FLOWS THROUGH THE 5 TIERS

1. Hardware Sensor Event
Driver speeds to 42 km/h with helmet ON. MPU6050 samples 1.0g motion. ESP32 packages JSON with Seq #10 and posts over WiFi with `X-Device-Key`.

2. Gatekeeper Ingestion
Express gateway checks `iotIngestLimiter`, verifies device auth, checks Zod bounds (0–300 km/h), deduplicates `eventId`, and normalizes clock.

3. PostgreSQL Persistence
Stored in `telemetry_readings` with `uq_telemetry_event_id` uniqueness. If impact ≥ 5.0g, creates provisional incident (Fault: UNKNOWN).

4. Actuarial Calculation
Engine normalizes events per 100 km, evaluates 5-factor model → Safety Score: 100/100 → Pricing Engine sets rate to ₹1,800/yr (10% discount).

5. Live Dashboard Delivery
React frontend polls every 4s, displaying live dials (42 km/h, ON), pulsing gateway clock badge, and the 100/100 Safe Driver card with ₹1,800 quote.